



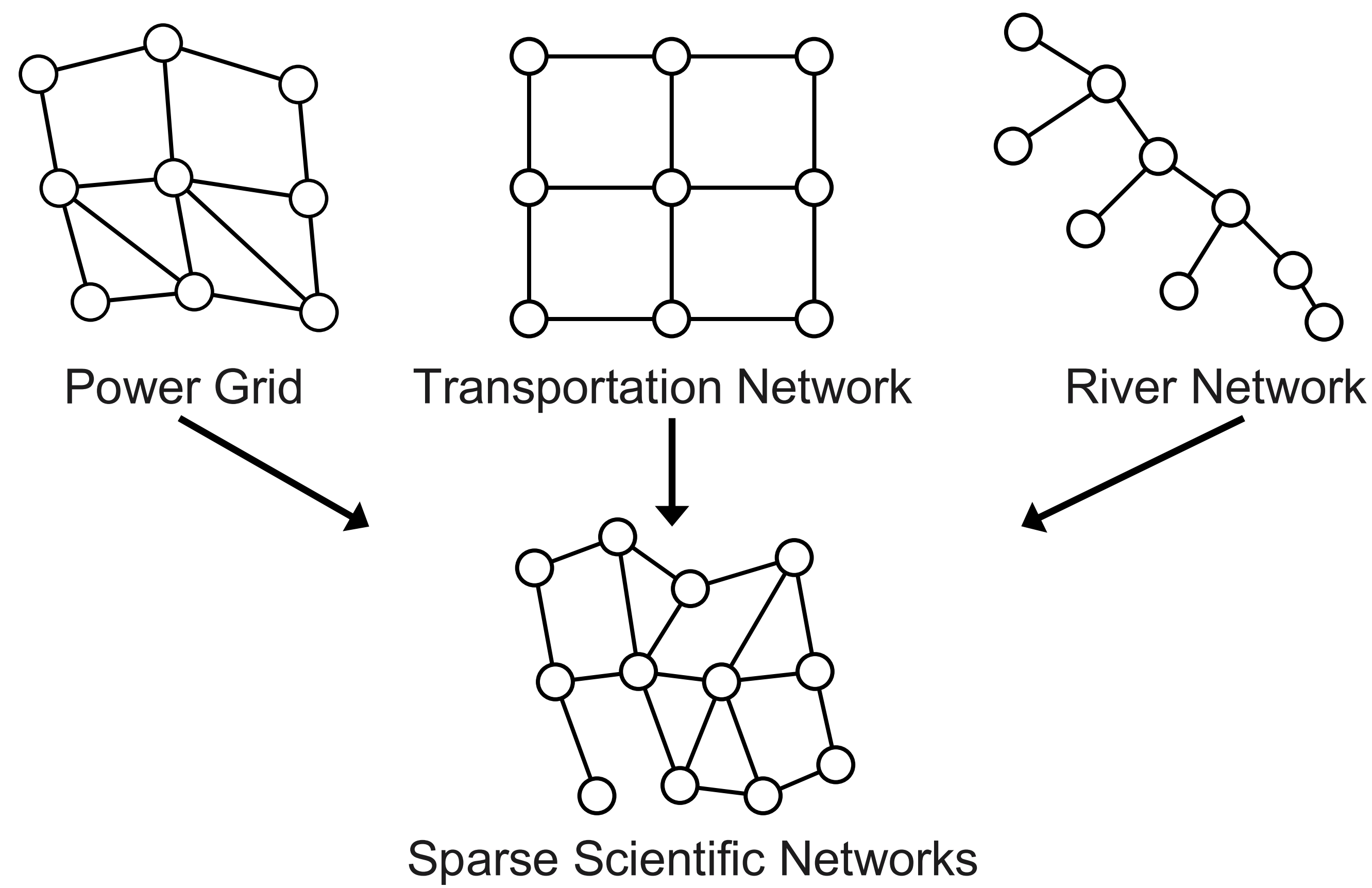
Robust Graph Learning for Sparse Scientific Networks

Developing mathematically grounded methods for trustworthy graph learning in sparse scientific networks



Pablo Moriano
Computer Science and Mathematics Division
Oak Ridge National Laboratory, Oak Ridge, TN 37830
moriano@ornl.gov

Scientific Motivation



Scientific Question

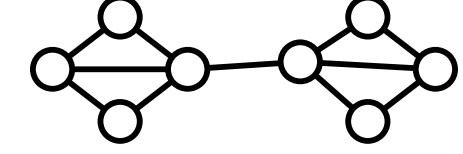
How reliable are graph learning methods on noisy, incomplete, manipulated, or evolving sparse scientific networks?

Scientific Framework

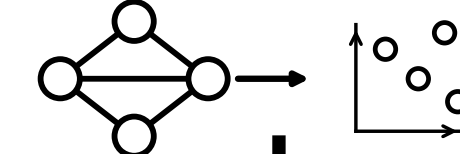
Scientific Hypothesis
Structural perturbations fundamentally influence the performance and reliability of graph learning

Methodological Families

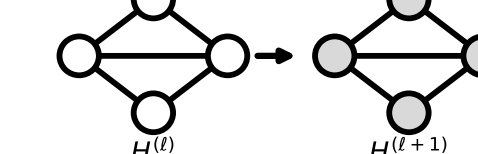
Network Analysis



Representation Learning



Graph Neural Networks

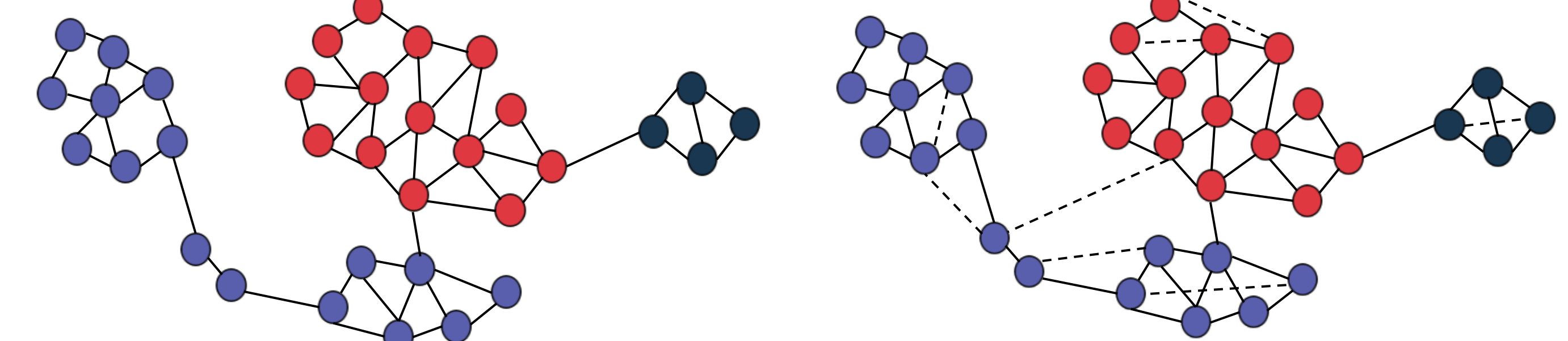


Toward
Trustworthy Graph Learning

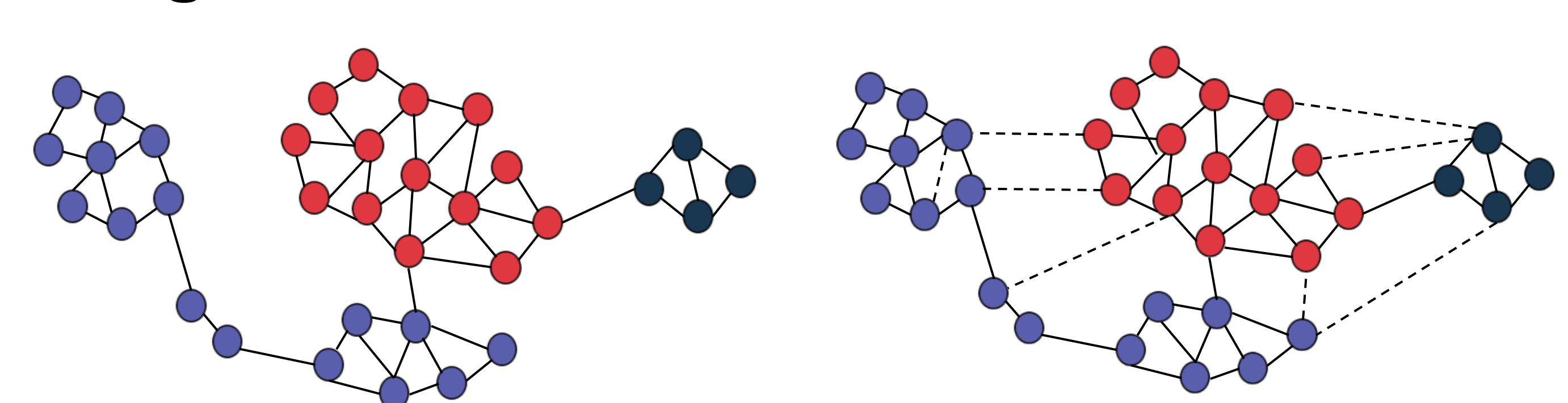
Network Analysis

How robust is community structure when sparse networks are perturbed?

• Random Perturbations



• Targeted Perturbations

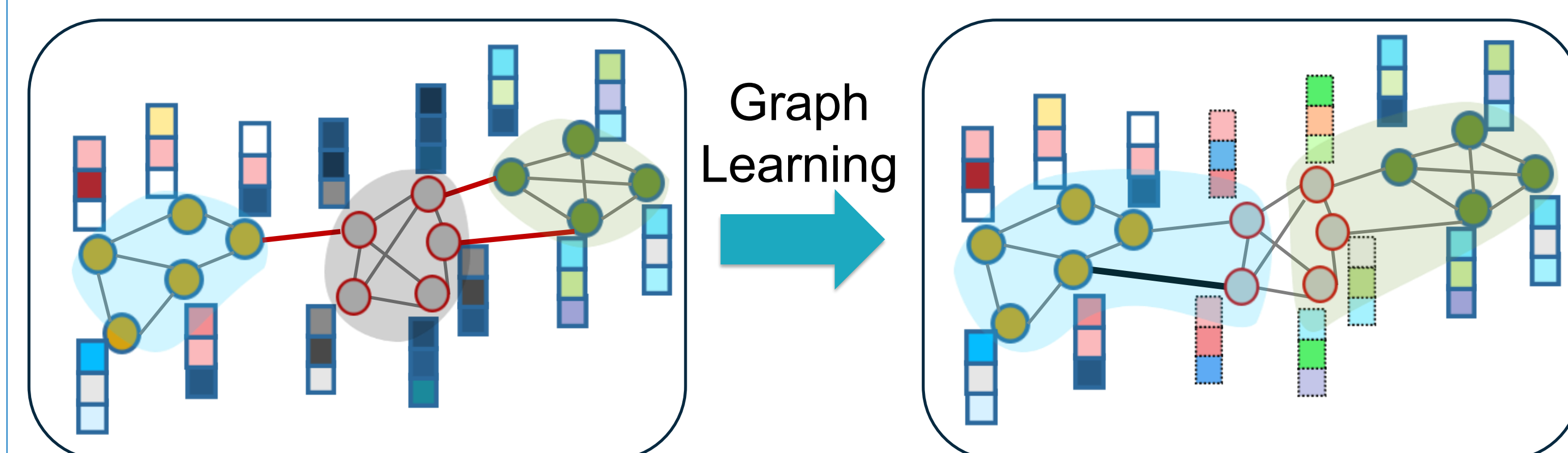
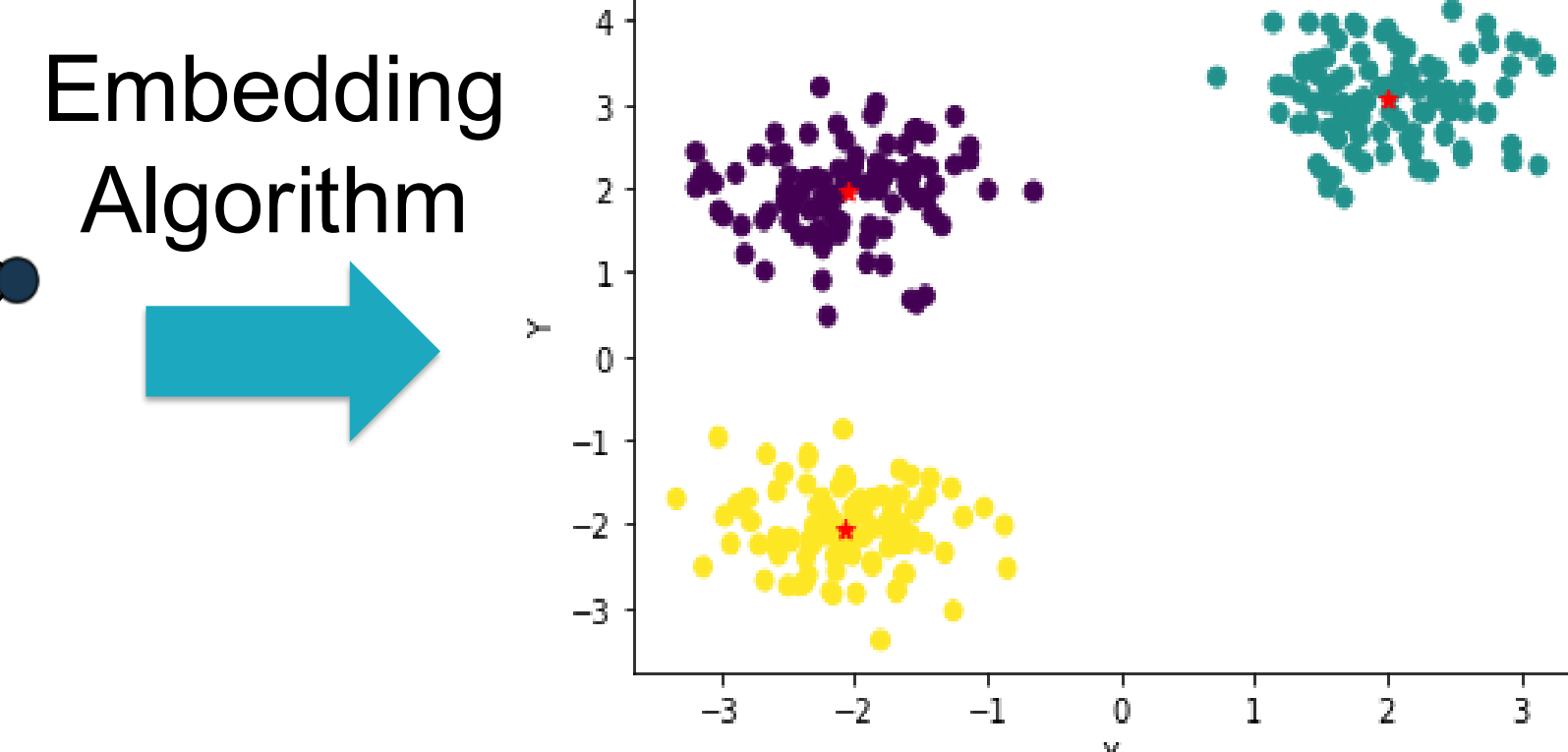


Key Finding

Robustness depends jointly on perturbation type, community strength, and detection algorithm

Representation Learning

How robust are graph representations to structural perturbations?



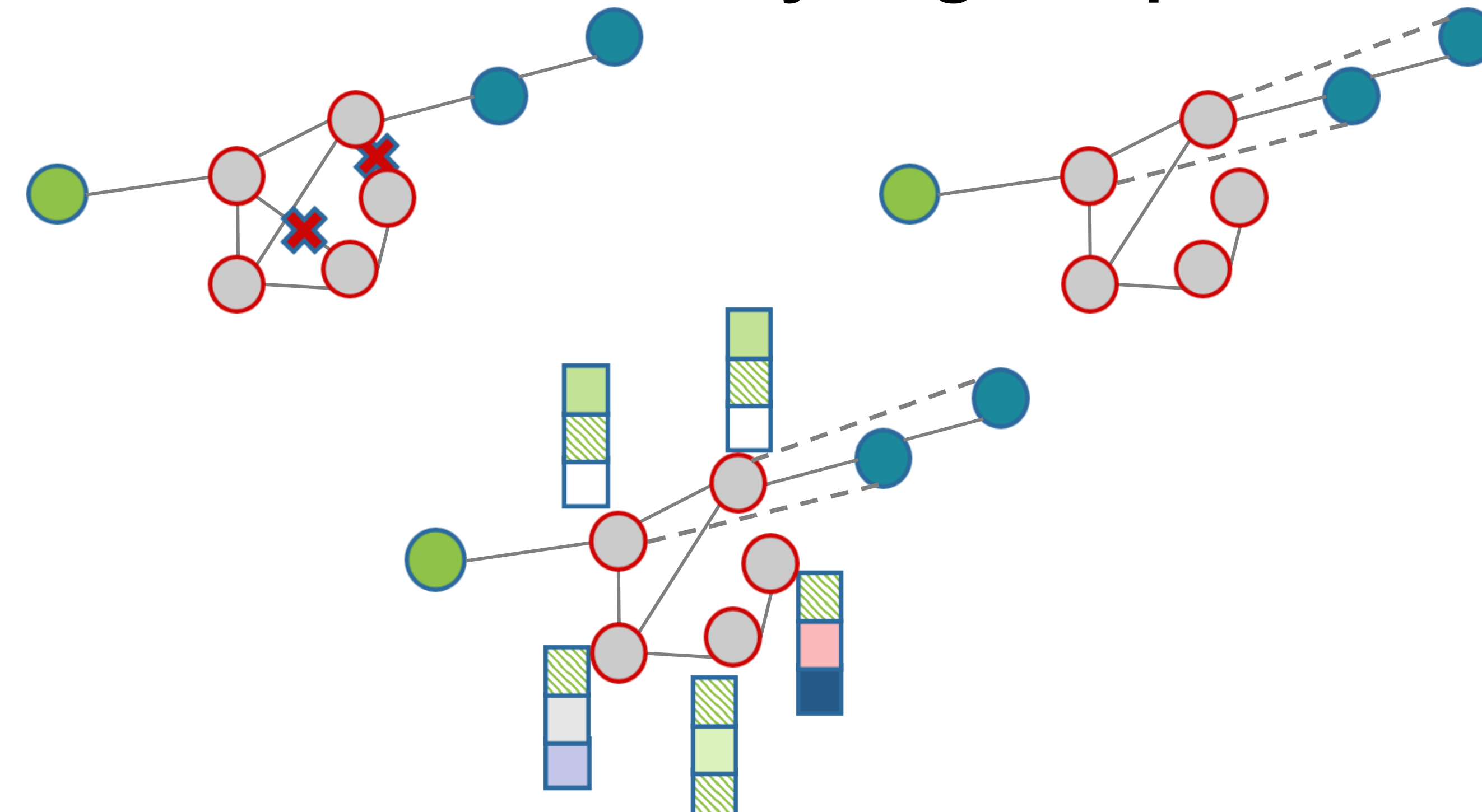
Key Finding

Embedding robustness is algorithm dependent and strongly influenced by graph structure, perturbation type, and embedding dimension

Community Hiding

Can robustness insights be translated into improved community hidability?

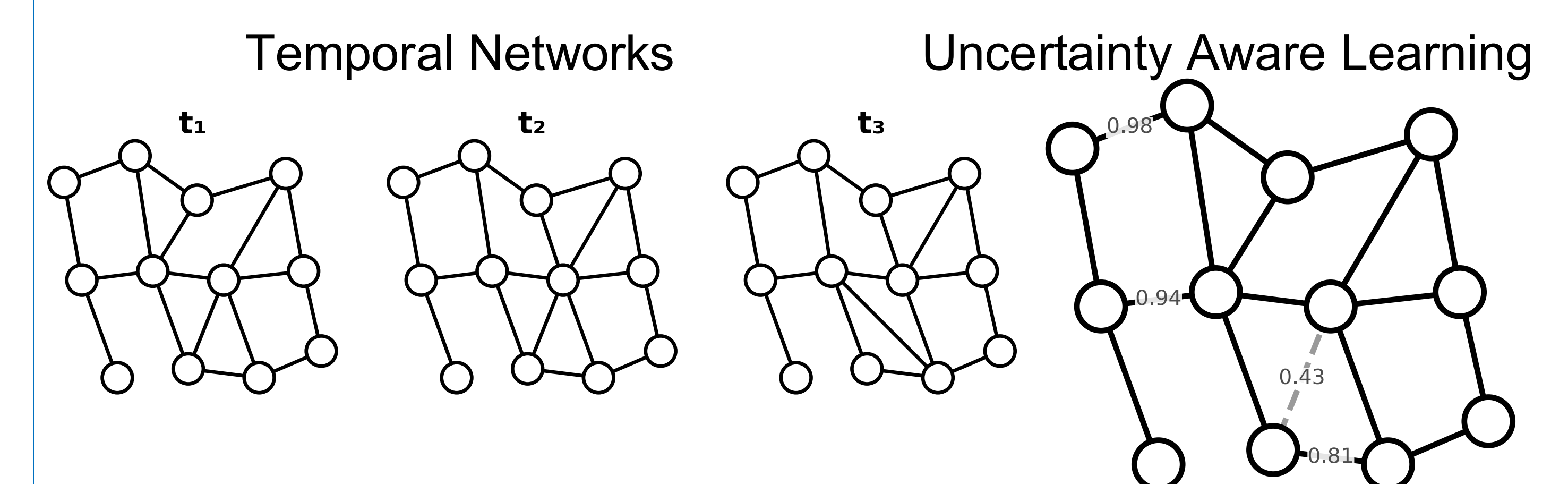
• F-ComDICE: Boundary targeted perturbation



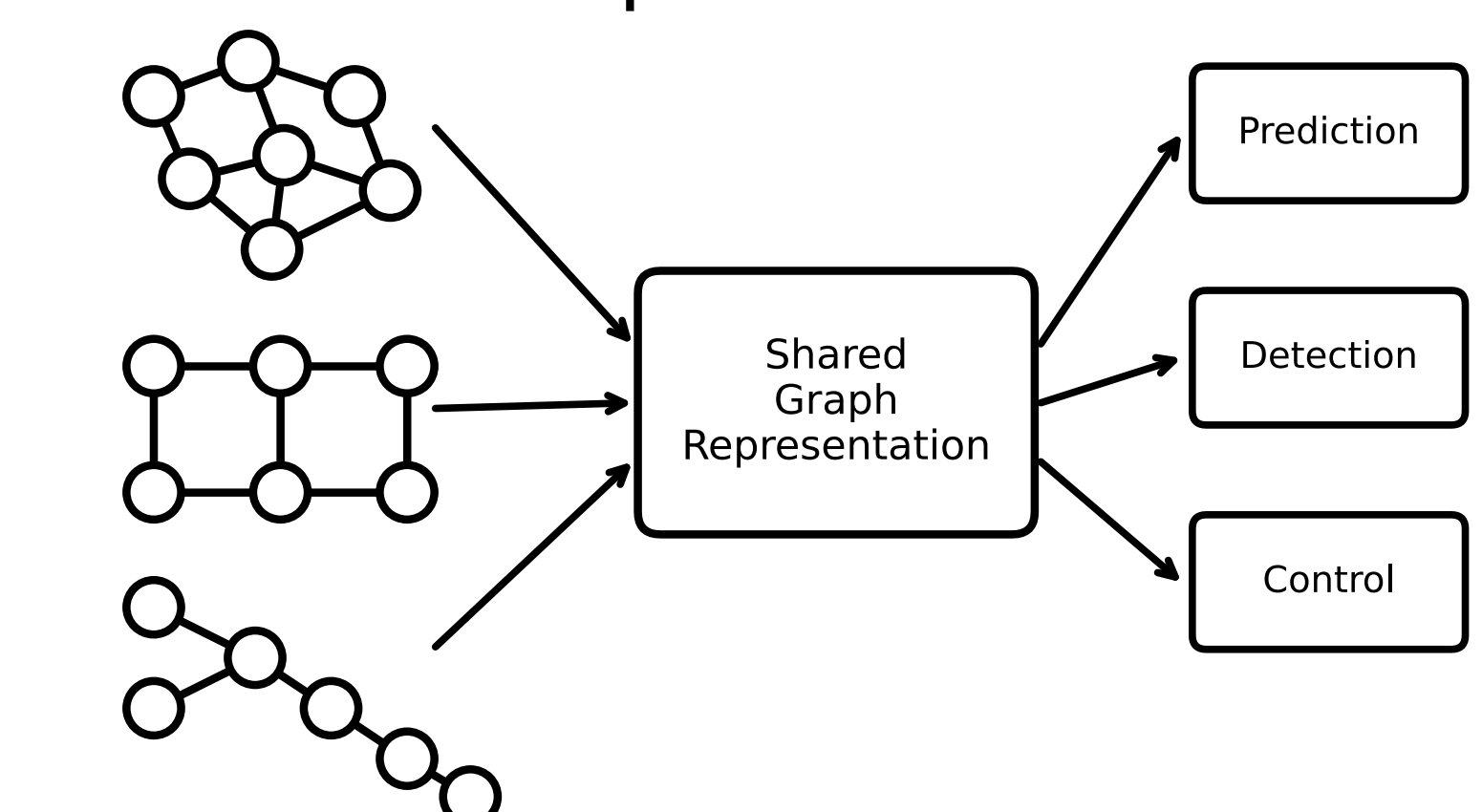
Key Finding

Understanding graph learning robustness enables the design of targeted structural perturbations that systematically improve community hidability

Future Research Directions



Scientific Graph Foundation Models



Vision

Advance trustworthy graph learning for dynamic, uncertain, and heterogeneous scientific networks

Supported by DOE SC MMICC program under Contract No. DE-AC05-00OR22725



Scan for Poster, Papers, and Code

Sparsitute: A Mathematical Institute for Sparse Computations in Science and Engineering, is part of the MMICCs program.
Partners: LBNL, ORNL, Purdue, Indiana, UIUC, Wake Forest



U.S. DEPARTMENT OF
ENERGY

Office of
Science

Pablo Moriano
Website: <https://pmoriano.com>
GitHub: <https://github.com/pmoriano>